# Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

Van Thuong Nguyen[a]

[a]The University of Danang, Viet Nam

ABSTRACT

Wheeled bipedal robots face a trade-off: the stiffness that buys precise standing shrinks the push-recovery envelope, while the integral action that removes equilibrium bias winds up during disturbances. The Anchored Conditionally-Gated Controller (ACC) answers both by *conditional activation*—a position integral confined to a 15 cm neighbourhood and enabled by an asymmetric envelope follower ($\tau_{\text{attack}}$≈23 ms, $\tau_{\text{release}}$≈1.5 s) that separates quiet stance from post-push ringdown. On a 10-DOF, 8.1 kg wheeled biped in MuJoCo (500 Hz physics, 100 Hz control) the architecture is dissected, not only demonstrated. A factorial ablation separates three coupled effects: gated damping produces the steadiness, the anchor integral the accuracy, and the gates are bit-identical to ACC in quiet stance yet decisive under disturbance. The push envelope is measurably chiral (−16.3 N), traced to the anti-symmetric hip-roll/hip-yaw channels. A closed-form model predicts the 27.0 mm standing offset to 0.92 % across a fortyfold gain sweep, as the price of a bounded integrator memory. ACC holds 0.294±0.015 mm sagittal repeatability ($N$=10), 59-fold over this ablation's proportional-only rung, recovers omnidirectional pushes (107 N median, 82 N worst case), and stands 30 min at 0.0004 mm/min drift where six classical baselines and a full-state LQR all fall. A 2 ms-resolved delay screen prices deployment at a 6 – 8 ms latency cliff that one retuned gain moves outward.

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## 1. Introduction

Wheeled biped robots need millimeter-level standing precision, omnidirectional push recovery, and terrain adaptation—but no single classical controller integrates all three: LQR handles balance (Klemm et al., 2019), WBC posture (Klemm et al., 2020), and MPC terrain (Wang et al., 2025); locomotion, separately again, through learned policies (Hwangbo et al., 2019).

The difficulty is fundamental. Lowering $k_p$ widens the capture basin at the cost of balance stiffness; omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a persistent, predominantly *sagittal* limit cycle regardless of $k_p$ ($N$=10: 69.7 mm peak-to-peak sagittally against 13.4 mm laterally), on the one axis the wheels can drive; yet a global integral winds up during pushes.

We present the Anchored Conditionally-Gated Controller (ACC), which addresses both dilemmas through *conditional activation*. The core mechanism is a **proximity-gated anchor**: a position integral confined to a 15 cm neighborhood, enabled by an **asymmetric envelope follower** ($\tau_{\text{attack}}$≈23 ms, $\tau_{\text{release}}$≈1.5 s) that distinguishes quiet stance from post-push ringdown. Coupled with a **gated damping boost**, ACC achieves 0.294±0.015 mm sagittal repeatability under idealized conditions (clean sensors, 0 ms delay, $N$=10), at push performance equal to or better than a proportional-only baseline carrying no integral action. Absolute accuracy is a separate quantity: the robot parks 27.0 mm from the commanded home and holds that standoff to 0.009 mm.

A factorial ablation under a single protocol (Table 1) separates three effects the architecture couples: quiet-stance *steadiness* comes from the gated *damping* terms rather than from the anchor integral, which buys *accuracy* instead; the proximity gate and the asymmetric envelope are provably inert during quiet stance, so only a displacement experiment can measure them; and the pitch safety gate is load-bearing with no disturbance applied. The gates are enabling conditions that make

CONTACT V. T. Nguyen nvthuong180702@gmail.com The University of Danang, Viet Nam.

an otherwise inadmissible damping and integral configuration safe, not the proximate source of the precision. The object of study is a *structured classical prior*: a hand-designed, interpretable torque assembly. Supplying the nominal base action of a residual-learning stack (Johannink et al., 2019) is the downstream use it was designed for and is future work; no claim below depends on a residual policy. Classical controllers are therefore the comparison of record: six of them—four 4-state TWIP and two 6-state coupled variants—all fall, and a full-state LQR linearized from the plant itself survives two orders of magnitude longer and still falls, so platform complexity rather than model order or servo lag dominates the failure. A preliminary model-free PPO (Schulman et al., 2017) run is a *difficulty probe* and bounds nothing about what RL attains here.

The main contributions are:

(1) A **proximity-gated anchor** with asymmetric envelope follower achieving sub-millimeter idle precision without sacrificing omnidirectional push survival.

(2) A **two-channel conditionally-gated torque assembly** whose channel separation admits contact-loss recovery and per-leg terrain adaptation as independent extensions without retuning the anchor; both are summarised here and specified in full in the extended version.

(3) A **factorial ablation** (additive L0→L3 + subtractive S0→S5) against six classical baselines, plus two control experiments that test the comparison itself: a full-state LQR derived from the plant rather than a reduced model, and a sweep of the coupled model's one empirical constant.

## 2. Related Work

### 2.1. Wheeled Bipedal Platforms and Classical Control

Ascento (Klemm et al., 2019) demonstrated two-wheeled jumping with LQR balancing, later extended to WBC with kinematic loops (Klemm et al., 2020); SKATER (Wang et al., 2025) uses hierarchical MPC with terrain adaptation; comparable platforms pair a balance controller with separate yaw, height and roll loops. Whleaper (Zhu et al., 2024) is closest to our morphology (10 DOF, three hip DOFs per leg); it *switches* between LQR for balanced sliding and a separately trained PPO policy for walking by locomotion mode rather than composing them on one control output. Composing a classical prior with a learned residual on one output is the residual-RL construction (Johannink et al., 2019), applied recently to wheeled-biped balance (Zhang et al., 2025); ACC supplies that prior. A recent review (X. Liu et al., 2024) notes the absence of a unified classical solution. Coupled pitch–roll LQR, standard on rigid two-wheeled platforms without articulated legs, reproduces on our morphology as a 6-state design and falls; so does the 4-state Direct Torque variant that removes the servo path, so actuation lag alone cannot account for the outcome (§5.4). Cascade control (Åström & Hägglund, 1995) and whole-body control via hierarchical inverse dynamics or task-priority QPs (Herzog et al., 2016) use fixed-gain nested or prioritized loops without conditional gates; gain scheduling (Rugh & Shamma, 2000) varies parameters continuously with one variable rather than responding to distinct physical regimes—the reason the task-priority QP we benchmark standalone and as an ACC assist layer (§5.5) does not resolve the precision–recovery trade-off. Split-range, selector and override architectures (Åström & Hägglund, 2006) blend continuously in process control; ACC's contribution is the synthesis of gating surfaces—spatial proximity, velocity envelope, pitch safety interlock—derived from wheeled-biped failure modes rather than generic process-variable limits.

### 2.2. Disturbance Rejection, Anti-Windup, and ACC's Distinction

DOB (Ohnishi et al., 1996) and ADRC (Han, 2009) estimate disturbances temporally but cannot distinguish a push from post-push ringdown—both occupy the same frequency band. Classical anti-windup—back-calculation, conditional integration, the unified observer-based framework (Åström & Rundqwist, 1989)—triggers on *actuator saturation*; ACC's anchor instead uses **spatial, physics-conditioned gating**, confining integration to a 15 cm neighborhood and separating quiet stance from ringdown by a velocity envelope, replacing the estimation-speed-versus-noise trade-off with an attack-versus-release one. Push recovery via capture point (Englsberger et al., 2015) and flight-phase control (Roscia et al., 2023) is complementary, and terrain adaptation is usually exteroceptive or learned (Lee et al., 2020); ACC extends proprioceptive contact-point estimation to wheeled bipeds. Hyon & Cheng (2006) made a humanoid passive to unmeasured external forces by gravity compensation with optimal contact-force distribution; ACC meets the same requirement by switching regime on proprioceptive gating variables rather than redistributing contact forces.

## 3. Robot Platform

### 3.1. Robot Morphology and Simulation

The robot (Fig. 1) is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). Mass: 8.1 kg; thigh: 0.26 m; shin: 0.28 m; wheel radius: 0.06 m; hip width: 0.23 m. Torque limits: 30 Nm (HR, HY, WH), 150 Nm (HP, KN). MuJoCo (Todorov et al., 2012) runs at 500 Hz ($\Delta t_{\text{sim}} = 0.002$ s); ACC and every classical baseline at 100 Hz (5 substeps), with the baselines additionally measured at their 50 Hz design rate and the PPO reference at the 50 Hz rate it was trained at. All experiments use a 400 Nm/s torque rate limit and raw torque commands direct to actuators. Integrator: implicitfast; cone: pyramidal (elliptic for wheels, $\mu_s$=1.2, $\mu_k$=0.8 body); armature 0.008 kg·m$^2$ (wheels) and 0.02 kg·m$^2$ (leg joints); contact `solref={0.002,1}` (body), {0.005,0.7} (wheels). Softening `solref[0]` to 0.02 (tire-on-asphalt) raises planar idle RMS by ~60 % to ~1.2 mm, so every sub-millimeter idle figure is conditional on stiff ground contact.

#### Height convention

**CoM-$z$**, the whole-body centre-of-mass height above ground, is what ACC commands and regulates; the nominal standing pose is $h_{\text{CoM}}$=0.404 m. **Base-$z$**, the torso root-body height (`qpos[2]`), sits 13.2 cm above it in that pose, varying by less than 1 cm over the postures used here. All ACC results are in CoM-$z$; the PPO reference (§5.4) and the WBC baselines (§5.5) command base-$z$ and are labelled accordingly, since substituting one convention for the other injects the full 13.2 cm offset as a reference error. ACC's posture map is calibrated at 0.354 m and 0.454 m and interpolated between them; outside that band joint targets are linearly extrapolated and bounded to [0.254, 0.554] m, the ±10 cm bound the per-leg height split (§4.5) operates against.

## 4. ACC Formulation

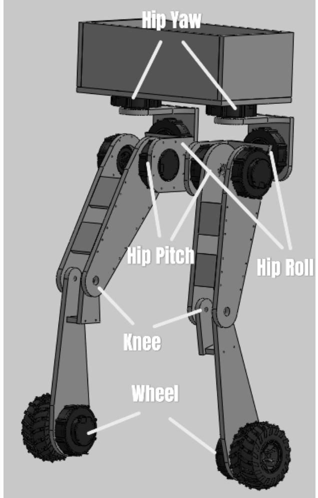


Figure 1. Robot joint layout with world frame (right-handed, $Z$ up). Wheel axles lie along world $X$, so the rolling—and therefore sagittal—direction is world $Y$ and world $X$ is the lateral (track) axis. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

### 4.1. Two-Channel Torque Assembly

ACC assembles a PD balance core, a proximity-gated anchor integral and posture regulation into two torque channels—wheel torque ($\boldsymbol{\tau}_w \in \mathbb{R}^2$) and leg-joint torque ($\boldsymbol{\tau}_q \in \mathbb{R}^8$)—with flight and terrain as independent extensions (Fig. 2):

$$\begin{aligned}\boldsymbol{\tau}_w &= \boldsymbol{\tau}_{\text{balance}} + g_{\text{anchor}}\boldsymbol{\tau}_{\text{anchor}} + g_{\text{flight}}\boldsymbol{\tau}_{\text{flight}},\\ \boldsymbol{\tau}_q &= \boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}}\Delta\boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd},L},\ h^{\text{cmd},R}).\end{aligned} \quad (1)$$

Every gate $g_*$ except the flight gate is a *smoothstep* function of a physical condition (proximity, quietness, attitude, terrain disparity) rather than a discrete switch:

$$\text{ss}(x; a, b) = 3t^2 - 2t^3, \qquad t = \text{clip}\left(\frac{x-a}{b-a},\ 0,\ 1\right). \quad (2)$$

Equation (2) is the standard Hermite blend, $C^1$ in $x$ with zero derivative at both knees. Continuity matters concretely: a hard threshold injects a torque step equal to the full gated component, which at 100 Hz under a 400 Nm/s rate limit reaches the plant as an unmodelled impulse. The one exception is $g_{\text{flight}}$, a discrete load-based hysteresis with asymmetric engage/release timers, because contact loss is a binary physical event.

Wheel torques control sagittal balance and yaw; leg-joint torques control posture and per-leg height adaptation. The two channels are coupled only through the plant—no term in $\boldsymbol{\tau}_w$ reads a leg-posture state and none in $\boldsymbol{\tau}_q$ reads the sagittal balance state—which is what allows flight recovery and terrain adaptation to be added as independent extensions without retuning the anchor. Unlike the industrial override architectures of §2.2, which select *among parallel controllers* on process-variable limits, ACC gates *components within a single torque assembly* on physical quantities of three kinds—spatial, temporal and safety-interlock—each chosen because a specific measured failure mode of this platform is expressible in it.

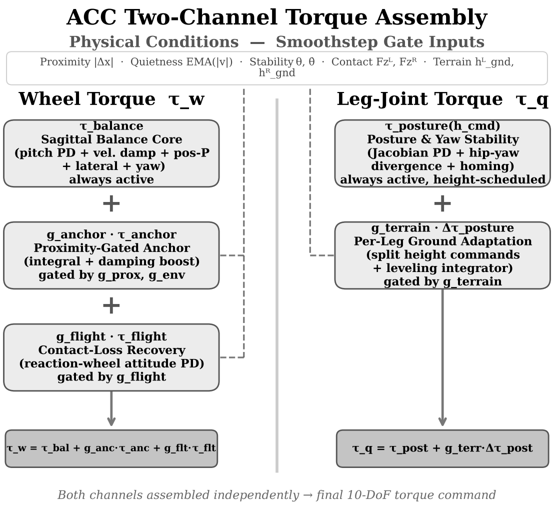


**Figure 2.** ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed).

## 4.2. *Sagittal Balance Core ($\tau_{balance}$)*

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\tau_{\text{balance}} = \begin{bmatrix} \tau_{\text{com}} + \tau_{\text{diff}} \\ \tau_{\text{com}} - \tau_{\text{diff}} \end{bmatrix}, \quad \begin{aligned} \tau_{\text{com}} &= \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}}, \\ \tau_{\text{diff}} &= \tau_{\text{lat}} + \tau_{\text{yaw}}, \end{aligned} \tag{3}$$

with common-mode and differential components

$$\tau_{\text{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \tag{4}$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \tag{5}$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x, \pm 0.10\,\text{m}), \tag{6}$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \tag{7}$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_{\psi} + k_{d,\text{yaw}} \cdot \omega_z, \tag{8}$$

where $\theta$ is torso pitch, $v_{\text{sag}}$ the sagittal body velocity, $\Delta x$ the sagittal position error from the latched home position, $\phi$ the roll angle and $e_{\psi}$ the yaw error.
The P position term ($k_{\text{pos}}$, clipped to $\pm 0.10$ m, saturating at $\pm 4$ Nm) keeps the robot within $\pm 4-6$ cm of home but leaves a 1.3 Nm equilibrium bias sustaining a perpetual limit cycle, which motivates the anchor integral. Saturation-triggered anti-windup does not address it, the critical failure being *state-dependent*: during a push, position error grows to $10-30$ cm while the wheels remain within their 20 rad/s range, so a saturation trigger never fires and ACC uses spatial confinement instead. Pitch stiffness is fixed at $k_p$=50; a scheduled variant softening it to 35 during pushes was tested and rejected (S0 vs. S1, Table 2).

## 4.3. *Proximity-Gated Anchor ($g_{anchor}\tau_{anchor}$)*

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\begin{aligned} \tau_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x)\, dt \\ + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}), \end{aligned} \tag{9}$$

where the gates are continuous smoothstep functions equal to 1 when the robot is "at home and quiet" and transitioning smoothly to 0 otherwise:

$$g_{\text{prox}}(\Delta x) = 1 - \text{ss}\big(|\Delta x|;\ 0.05\,\text{m},\ 0.15\,\text{m}\big), \tag{10}$$

$$g_{\text{env}}(v_{\text{sag}}) = 1 - \text{ss}\big(\text{EMA}(|v_{\text{sag}}|);\ 0.25\,\text{m/s},\ 0.50\,\text{m/s}\big), \tag{11}$$

$$\begin{aligned} g_{\theta}(\theta, \dot{\theta}) = \big[1 - \text{ss}(|\theta|;\ 2°,\ 12°)\big] \\ \times \big[1 - \text{ss}(|\dot{\theta}|;\ 2\,°/\text{s},\ 15\,°/\text{s})\big], \end{aligned} \tag{12}$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_{\theta}. \tag{13}$$

Each gate is dimensionless, monotone and $C^1$, and three physically distinct conditions must hold simultaneously for the anchor to integrate: spatially near home ($g_{\text{prox}}$), temporally quiet ($g_{\text{env}}$) and attitudinally settled ($g_{\theta}$). The pitch gate closes when either pitch or pitch rate leaves its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope.
**Asymmetric envelope follower.** Reliable quiet-stance detection rests on an asymmetric exponentially-weighted moving average (EMA) of $|v_{\text{sag}}|$:

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1-\alpha_a)\text{EMA}_{t-1}, & |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1-\alpha_r)\text{EMA}_{t-1}, & \text{else}, \end{cases} \tag{14}$$

with time constants $\tau_{\text{attack}}{\approx}23$ ms and $\tau_{\text{release}}{\approx}1.5$ s as the primary specification, the discrete coefficients following as $\alpha_a{=}1{-}e^{-\Delta t/\tau_a} \approx 0.35$ and $\alpha_r \approx 0.007$ at 100 Hz. The asymmetry is essential: a symmetric EMA is *bistable*—post-push oscillation holds it in a middle band where the gate is half-open and the boost cannot collapse the cycle, as the symmetric-EMA ablation of §5 confirms—whereas the fast-attack/slow-release envelope rises to 1 only when oscillation truly decays and drops to 0 within ∼25 ms of a disturbance. The mechanism is a peak-hold envelope detector; ACC's contribution is coupling it to a spatial proximity gate for integral conditioning. Since $\mathrm{EMA}_0$=0 biases $g_{\mathrm{env}}\approx 1$ for ∼7.5 s at startup, all measurements discard a 3 – 5 s settle interval.

### 4.4. Posture and Yaw Stability ($\tau_{posture}$)

Posture regulation uses Jacobian-based PD gains scheduled across 23 heights:

$$\tau_{\mathrm{posture}} = \tau^{\mathrm{jac}}_{\mathrm{PD}}(q, \dot{q}, h^{\mathrm{cmd}}) + \tau_{\mathrm{div}}(q^L_{\mathrm{hy}}, q^R_{\mathrm{hy}}) + g_{\mathrm{settled}}\,\tau_{\mathrm{homing}}(q - q_{\mathrm{ref}}), \tag{15}$$

where $\tau_{\mathrm{div}}$ damps hip-yaw divergence and $\tau_{\mathrm{homing}}$ restores nominal posture when upright and settled. Yaw return uses wheel-differential torque.

### 4.5. Extensions: Contact Loss and Per-Leg Terrain

Two extensions attach to the same assembly without re-tuning the anchor, and both vanish identically in nominal standing. *Contact-loss recovery* ($g_{\mathrm{flight}}$): when both wheels report $F_z < 0.5mg$ for two consecutive steps, ACC substitutes a flight attitude controller that uses the wheels as reaction flywheels, $\tau_{\mathrm{flight}} = \mathrm{clip}(2.0\theta + 0.5\dot{\theta} - 0.02\omega_{\mathrm{wheel}}, \pm 2\,\mathrm{Nm})$, with a five-step loaded-release hysteresis so balance torque cannot re-engage mid-bounce. *Per-leg ground adaptation* ($g_{\mathrm{terrain}}$): a wheel resting at a different ground height would otherwise force the torso to roll by the step angle, so ACC maintains an independent ground estimate per leg from the wheel contact-point height—exact at any body tilt for a rigid wheel—updated only while that wheel carries at least $0.2mg$, and absorbs the disparity by a *kinematic* leg-length split plus a closed-loop levelling integrator trimming the residual by the measured roll. Both are specified in full, with their ablation matrices, in the extended version.

### 4.6. Complete Numerical Parameter Set

The core gains are $k_p$=50 Nm/rad, $k_d$=3 Nm/(rad/s), $k_{\mathrm{pos}}$=40 Nm/m (clipped at ±4 Nm), $k_{\mathrm{vel}}$=15 Nm/(m/s) at damping scale 1.5, $k_{\mathrm{roll}}$=55 Nm/rad, $k_{\mathrm{yaw}}$=1.0 Nm/rad; the anchor uses $K_i$=4.0 Nm/(m s) capped at 2 Nm with a $5 \times 10^{-3}$ per-step leak, boost scale 5.0, $g_{\mathrm{prox}}$ band 0.05 – 0.15 m, $g_{\mathrm{env}}$ band 0.25 – 0.50 m/s, $g_{\mathrm{stab}}$ bands 2 – 12° and 2 – 15 ° s$^{-1}$; flight uses P/D = 2.0/0.5 with 0.02 Nm/(rad/s) wheel damping and 2/5-step engage/release at $F_z$=0.5$mg$ (0.2$mg$ for terrain). The full set (50+ gains) is in the supplementary material.

## 5. Experimental Validation

ACC and all six classical baselines are evaluated at 500 Hz physics / 100 Hz control under a 400 Nm/s per-joint torque rate limit, so the headline comparison is rate-matched by construction; the PPO reference is reported at the 50 Hz rate its policy was trained at. Each classical baseline was additionally measured at its own 50 Hz design rate: the fall rate is 100 % in all twelve cells and five of the six survive *longer* at 50 Hz, so reporting them at 100 Hz is the rate-matched choice and the harsher of the two for them. Seeds are deterministic: 20260727+trial_id for idle standing, 20260728×100+rep for the push ablation, and {0, 42, 123} for the classical baselines.

### 5.1. Anchored Standing

#### 5.1.1. Metrics and axis convention

"Standing precision" names two quantities that are not interchangeable. With $\mathbf{p}(t) \in \mathbb{R}^2$ the horizontal base position over a window of length $T$, $\mathbf{p}_0$ the commanded home and $\bar{\mathbf{p}}$ the trial mean,

$$A = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \mathbf{p}_0\|^2\,dt\right)^{1/2}, \tag{16}$$

$$R = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \bar{\mathbf{p}}\|^2\,dt\right)^{1/2} \tag{17}$$

define the *accuracy* $A$, which retains any DC parking offset, and the *repeatability* $R$, which removes it; the two differ by more than an order of magnitude here, so both are reported for every configuration. Each is resolved along the sagittal axis, the lateral axis and the full plane, the two horizontal axes being actuated by different means: the **sagittal** axis is world $y$, the rolling direction and the only one on which wheel torque produces ground force, while the **lateral** axis is world $x$, held passively by the wheel contacts and the roll restoring moment of the track width. A dynamic check confirms the assignment (`scripts/verify_body_axes.py`): a 90 N impulse along $y$ is absorbed at a peak wheel speed of 52 rad/s leaving 4.8 mm of residual offset, whereas the same impulse along $x$ leaves a permanent 23.2 mm offset. The anchor integral and the gated damping boost both live in the wheel-torque channel, so both act sagittally only.

#### 5.1.2. Protocol

Every row of Table 1 was re-measured under one protocol: 25 s from the nominal standing pose with randomised initial conditions ($\sigma$=0.005 rad per joint, clipped to limits; $\sigma$=1 mm on root height), with $A$, $R$ taken relative to each trial's own $t$=0 pose after a settle covering the EMA$_0$=0 transient to within 1 %. A trial is a fall if $|\theta|>46°$ or root height <0.15 m; confidence intervals are Student-$t$. This measures the ability to *stay* put; coming back is measured in Sec. 5.2.

#### 5.1.3. Result

ACC holds **sub-millimetre sagittal repeatability**, $R_{\text{sag}}$ = 0.294±0.015 mm (95% CI [0.283, 0.305] mm) across ten randomised trials with no falls—the figure of merit for the platform, sagittal being the actuated inverted-pendulum axis. Precision is assessed from the relative spread *between* configurations under an identical protocol rather than against a simulator noise floor: removing a single mechanism (S2) moves $R_{\text{sag}}$ sixteenfold with non-overlapping SDs. Against the P-only baseline L0, which sustains a 69.7 mm peak-to-peak sagittal limit cycle, the like-for-like improvements are **59.4**× sagittal, **22.6**× planar and **5.0**× lateral—largest on the hard axis and smallest on the easy one, the correct signature for a wheel-torque controller, since every term ACC adds beyond L0 produces ground force along the rolling direction and none can produce lateral force at all.

Accuracy and repeatability are distinct on this axis. ACC parks at $\bar{s}$ = −27.0 mm from commanded home and holds that standoff to 0.009 mm SD across trials, so planar accuracy is $A_{xy}$ = 26.994 mm, a 1.8× improvement over L0 against the 59× gain in repeatability. Its magnitude is fully accounted for: instrumenting the sagittal torque balance predicts the offset in closed form from a *leaky* anchor integral whose ~2 s forgetting horizon fixes its DC gain at the finite $k_{I,\text{dc}}$=7.96 Nm/m—so a constant load leaves a finite error by construction—and a 1.44° error in the feedforward pitch reference, giving 24.88 mm against 24.99 mm measured and holding across a 40× gain sweep and a five-point height sweep to within 0.92 %. The leak also buys the steadiness: a 10× smaller $\lambda$ cuts the offset to −16.31 mm but degrades window jitter twelvefold, and Table 1 publishes a deliberately chosen point on that trade-off.

#### 5.1.4. The measurement window is a protocol choice, not a stability horizon

Every row of Table 1 is measured over 20 s; §5.4.1 declines to promote a competing design because that horizon flatters it, and the standard cuts both ways. Re-running the ACC row—identical seeds, initial conditions, profile and metrics, only the window lengthened—reproduces 0.294±0.015 mm at 20 s and then *improves* on it, to 0.089±0.005 mm at 300 s (10/10, drift +0.0149 mm/min) and 0.037±0.002 mm at 1800 s (4/4, +0.0004 mm/min), so the 20 s figure is the conservative one: that window still contains the tail of the settle. The settle terminates in a fixed point, not a slow mode—after ≈35 s the sagittal coordinate is constant to $2 \times 10^{-6}$ mm per 30 s block for the remaining twenty-nine minutes, the parking offset holds at −26.763 mm, and the only motion left in the state is a lateral creep of 0.14 mm/hour on the passively held axis. The separation from the full-state LQR of Section 5.4.1, which accumulates 0.43 – 0.63 m of planar drift and falls, is therefore qualitative: ACC's position error is referenced to a latched home by two explicit terms, the clipped proportional $\tau_{\text{pos}}$ of Eq. (3) and the anchor integral on $-\Delta x$ of Eq. (9).[1]

#### 5.1.5. Which mechanism produces the precision

The lower blocks of Table 1 isolate the contributions; three of the four findings contradict the attribution the gate structure invites. *(i) Idle steadiness comes from the damping ladder; the integral buys accuracy instead.* M1 disables the anchor integral ($K_i$=0) with every gate in place, and its repeatability is slightly better than ACC's while its accuracy is worse by 4.2 mm; M2 keeps the integral and every gate but reverts the two velocity-damping coefficients to their L2 values, reproducing the L2/L2.5 limit cycle exactly. Steadiness is therefore attributable to the damping terms, in two stages: raising the velocity-damping scale takes $R_{\text{sag}}$ from the L2 limit cycle to 3.97 mm (M3) and the gated boost takes it the rest of the way, a combined factor of 86. *(ii) The damping boost acts sagittally, as it must.* S2 degrades sagittal repeatability sixteenfold while leaving lateral repeatability statistically unchanged (overlapping CIs), the only outcome consistent with a boost inside the wheel-torque channel.

*(iii) The proximity gate and the asymmetric envelope are*

[1] Seeds 20260727+$i$, 5 s settle, clean sensors, 0 ms delay, as in Table 1; produced by `scripts/idle_longhorizon.py`.

**Table 1.** Idle standing precision across the full ablation ladder. All rows re-measured under the single protocol of Sec. 5.1; values are mean±SD over $N$=10 trials.

| | Configuration | $R_{\text{sag}}$ (mm) sagittal rep. | $R_{xy}$ (mm) planar rep. | $R_{\text{lat}}$ (mm) lateral rep. | $\bar{s}$ (mm) sag. offset | Survival |
|---|---|---|---|---|---|---|
| **Additive ladder (bottom-up)** | | | | | | |
| L0 | P-only (PD, no position, no integral) | $17.461 \pm 0.258$ | $17.845 \pm 0.279$ | $3.675 \pm 0.246$ | −44.8 | 10/10 |
| L1 | + position homing (±0.10 m) | $20.397 \pm 0.017$ | $20.414 \pm 0.019$ | $0.834 \pm 0.066$ | −30.4 | 10/10 |
| L2 | + ungated integral $K_i$ | $25.026 \pm 0.039$ | $25.056 \pm 0.043$ | $1.212 \pm 0.110$ | −27.5 | 10/10 |
| L2.5 | + proximity gate only (no envelope, no boost) | $25.198 \pm 0.017$ | $25.227 \pm 0.020$ | $1.205 \pm 0.099$ | −27.6 | 10/10 |
| **Proposed controller** | | | | | | |
| L3/S1 | **ACC** (full anchor, fixed $k_p$=50) | $\mathbf{0.294 \pm 0.015}$ | $\mathbf{0.788 \pm 0.069}$ | $\mathbf{0.730 \pm 0.073}$ | **−27.0** | **10/10** |
| **Subtractive ablation (top-down from ACC)** | | | | | | |
| S2 | − gated damping boost | $4.772 \pm 0.069$ | $4.824 \pm 0.067$ | $0.703 \pm 0.069$ | −26.8 | 10/10 |
| S3 | − asymmetric envelope (symmetric EMA) | $0.294 \pm 0.015$ | $0.788 \pm 0.069$ | $0.730 \pm 0.073$ | −27.0 | 10/10 |
| S4 | − proximity gate (integral always on) | $0.294 \pm 0.015$ | $0.788 \pm 0.069$ | $0.730 \pm 0.073$ | −27.0 | 10/10 |
| S5 | − pitch safety gate $g_\theta$ | — | — | — | — | **0/10** |
| **Single-mechanism isolation (ACC with one term disabled)** | | | | | | |
| M1 | $K_i$=0 (anchor integral removed, gates intact) | $0.089 \pm 0.035$ | $0.733 \pm 0.072$ | $0.727 \pm 0.071$ | −31.2 | 10/10 |
| M2 | damping terms reverted to their L2 values | $25.198 \pm 0.017$ | $25.227 \pm 0.020$ | $1.205 \pm 0.099$ | −27.6 | 10/10 |
| M3 | $K_i$=0 *and* boost = 0 | $3.973 \pm 0.130$ | $4.037 \pm 0.128$ | $0.710 \pm 0.069$ | −31.1 | 10/10 |

100 Hz control, 20 s window after a 5 s settle, clean sensors, 0 ms delay. $R$ = repeatability (Eq. (17)); $\bar{s}$ = mean signed sagittal parking offset; accuracy on an axis is that offset and the repeatability in quadrature, so the $A$ columns are omitted (the identity holds to 0.003 mm on every surviving row). Sagittal is world $y$, lateral world $x$; the anchor integral and the damping boost act sagittally only, so $R_{\text{sag}}$ and $\bar{s}$ discriminate between configurations while $R_{\text{lat}}$ measures a passively held axis. Produced by `scripts/collect_idle_ladder.py`.

*inert during quiet stance.* Ablating either (S4, S3) is *bit-identical* to ACC on every idle metric—the base never leaves the 5 cm inner band and the velocity envelope never rises into the attack branch—so both are correctly evaluated only in Table 2. *(iv) The pitch safety gate is required even to stand.* S5 falls in 10/10 trials during quiet stance with no push applied: removing $g_\theta$ leaves the damping boost free to act on the pitch state at large excursions, and the resulting positive feedback diverges from the ±0.005 rad initial perturbation alone. The anchor's *gates* therefore make an aggressive damping and integral configuration admissible without windup or divergence; the damping terms they admit deliver the steadiness.

## *5.2. Omnidirectional Push Recovery*

Push recovery is evaluated by polar sweep: 10 – 160 N impulses of 7 control steps applied at the torso in 8 world-frame directions ($\theta = \pm 90°$ sagittal, $\theta = 0°/180°$ lateral), each threshold located by binary search to 5 N, survival being torso pitch < 46° for 17 s post-push; $F_{\min}$ is the maximum magnitude survived in *every* direction and $F_{\text{med}}$ the median.

*The envelope is anisotropic by construction.* Per-bearing means over $N$=10 repetitions span 82.3 – 144.8 N: the two lateral bearings average 98.2 N against 131.3 N for the two sagittal ones, and the weakest bearing of all is the oblique 45° (82.3 N, 23 % below $F_{\text{med}}$). The asymmetry follows from the morphology rather than from tuning: the wheels can drive the contact patch back under the CoM sagittally but produce no lateral ground force, so raising the lateral bearing requires a centralised lateral task (§5.5), not a gain change.

*The envelope is also chiral, and the chirality is in the controller.* Superposed on the anisotropy is a left–right asymmetry: 135° survives 40.3 N more than −135°. The plant cannot produce it—mirroring the model about the sagittal plane reproduces its mass, inertia and geometry to 0.05 mm and $1 \times 10^{-7}$—and a harmonic fit localises the violation in the mirror-forbidden term ($b_2$=−16.3 N against $b_1$=0.3 N). Under a mirrored state the commanded torque maps to its own mirror image exactly on hip-pitch, knee and wheel and fails only on hip-roll and hip-yaw, both assembled antisymmetrically; mirroring those two channels flips $b_2$ to 15.6 N at unchanged magnitude while a sham mirror leaves it at −15.7 N. The effect is reported without adopting the correction, which redistributes the envelope without enlarging it ($F_{\min}$ 82.5→79.7 N). Later protocols apply a single impulse along world +$x$, a lateral and below-median bearing (95.6 N), so those experiments are conservative probes.

Table 2 reports the factorial ablation. Additively, only homing (L1) and the full anchor (L3) add significant

**Table 2.** Factorial ablation of ACC components: push-recovery and ringdown metrics. Clean sensors, all rows $N$=10, mean±std.

| | Configuration | $F_{\min}$ (N) | $F_{\text{med}}$ (N) | Ringdown |
|---|---|---|---|---|
| **Additive (bottom-up)** | | | | |
| L0 | P-only (PD, no pos., no I) | 70.0±1.0 | 122.1±2.1 | never |
| L1 | +Pos. homing (±0.10 m) | 74.8±0.8 | 130.5±1.2 | never |
| L2 | +Integral $K_i$ (no gate) | 74.8±1.0 | 131.1±1.1 | — |
| L2.5 | +Prox. gate only (no EMA, no boost) | 75.3±0.8 | 131.9±2.4 | — |
| L3 | +Full anchor (gate, EMA, boost) | **82.3**±**2.2** | 106.7±7.0 | **9.0** s |
| **Subtractive (top-down from ACC)** | | | | |
| S0 | ACC + scheduled $k_p$ (rejected) | 81.0±2.2 | 109.2±7.6 | 9.0 s |
| S1 | **ACC** (canonical, fixed $k_p$=50) | **82.3**±**2.2** | 106.7±7.0 | **9.0** s |
| S2 | −Damping boost | 80.8±0.9 | 116.2±5.3 | never |
| S3 | −Asym. EMA (sym. EMA) | 81.6±1.8 | 106.2±6.7 | never |
| S4 | −Prox. gate (global I) | 80.5±2.7 | 109.2±7.6 | windup |
| S5 | −Pitch gate $g_\theta$ | <10 | <10 | never |

$F_{\min}$ and $F_{\text{med}}$ are the minimum and median survivable push over 8 bearings, each located by binary search to 5 N tolerance; idle metrics for the same configurations are in Table 1. The S5 entries are left-censored at the search floor: in all 80 trials the bisection never located a survivable magnitude in [10 N, 160 N]. Welch's unpaired two-sided $t$-test ($N$=10 per group, Bonferroni-adjusted $\alpha$=0.007): L0→L1 $p<1\times10^{-9}$, $d$=5.4; L1→L2 $p$=0.88; L2→L2.5 $p$=0.19; L2.5→L3 $p<2\times10^{-6}$, $d$=4.1. No subtractive row differs significantly from ACC except S5 ($p<1\times10^{-14}$). At $N$=10 the 95% CI half-width is ≈0.72×std, so sub-2 N differences are not resolvable. Each trial starts from a freshly constructed controller context. Produced by `scripts/replicate_ablation_n10.py`.

margin; the global integral and the proximity gate contribute nothing to push *margin* on their own—the gate's role is windup prevention, not capture-basin enlargement—and the full anchor is the only ladder configuration that settles. Subtractively the ablation is flat: every single-mechanism removal except the pitch gate leaves $F_{\min}$ within 1.8 N of ACC and none is significant, so the capture basin is set by the anchor as a whole rather than any one of its four gates—including gain scheduling (S0) and the damping boost, which dominates idle precision yet leaves push margin indistinguishable from ACC's (S2). The one mechanism the push data *do* isolate is the pitch gate: removing it (S5) leaves the robot unable to survive even the search floor. Post-push ringdown decays within 9 s under ACC, whereas P-only oscillates indefinitely because the 1.3 Nm equilibrium bias sustains the limit cycle; the symmetric-EMA ablation exhibits the bistable failure the asymmetry is designed to prevent, and the ungated integral winds up during the push, preventing settling entirely.

### *5.3. Flight Recovery and Terrain Adaptation*

At $N$=10 randomised-IC trials, ACC settles 10/10 from every drop and ledge tested—peak $|\theta|$ of 24.0±1.2° on a 100 cm drop and 27.1±2.0° on a 50 cm ledge—and traverses 10/10 at every curb height to 20 cm (0.15 m/s) at a straddle roll of 2.4±0.1° to 3.9±0.1°. Replacing the per-leg split with a symmetric posture falls at *every* curb height. The full ablation matrices are reported in the extended version.

### *5.4. Comparison with Classical Baselines and a Preliminary RL Reference*

To establish that ACC is not merely a tuned linear controller we compare six classical baselines under identical environment setups: four derived from the 4-state TWIP model (Grasser et al., 2002) (LQR with decoupled roll/yaw PD through a PID servo layer of ∼0.25 s lag; the same plus a pitch-error integral with conditional anti-windup; a direct-torque variant re-optimised for the zero-lag plant; and PI+AW, adding a conditional forward-position integral) and two 6-state coupled variants adding the roll states $(\phi, \dot{\phi})$, one through the servo and one commanding torque directly, all weights following Bryson's rule (Bryson & Ho, 1975). A model-free PPO run (5.4M steps, single seed, no domain randomisation or hyperparameter search) is reported alongside as a difficulty probe, not a tuned RL baseline (Limitation 4).

All six classical baselines fall within 0.60 s. Removing the PID servo path cuts torque 2.5× and nearly doubles survival but *worsens* roll: the recovered authority is spent in the sagittal plane the model describes, not in the lateral one that ends the episode. A negative result is only as strong as its tuning, so the family is swept rather than tuned once. A 46-configuration retuning sweep leaves the outcome unchanged, as does an 80× sweep of the coupled model's one empirical constant, the hip-roll-to-roll-acceleration coupling $b_{r,h}$: the servo variant falls at 0.320 s for every value, roll RMS spread 0.06 ‰, though the re-derived roll gain varies 32×. That is invariance to the parameter, not insensitivity of the metric; the mechanism is saturation—the roll channel is commanded against the ±0.7 rad hip-roll limit throughout every episode. The preliminary PPO

**Table 3.** Classical baselines and the preliminary PPO reference vs. ACC, clean sensors. All rows at 100 Hz control except the PPO reference, which is reported at the 50 Hz rate its policy was trained at.

| Configuration | Surv. (s) | $\theta_{RMS}$ (°) | $\phi_{RMS}$ (°) | $H_{RMS}$ (mm) | $\tau_{RMS}$ (Nm) |
|---|---|---|---|---|---|
| LQR, TWIP ($N$=20) | 0.32 | 0.006 | 24.2 | 205.8 | 21.99 |
| LQR + AW ($N$=20) | 0.32 | 0.006 | 24.2 | 205.8 | 21.99 |
| LQR, DT (re-opt.) ($N$=20) | 0.60 | 0.033 | 27.5 | 119.2 | 8.76 |
| 6-St. LQR ($N$=20) | 0.32 | 0.006 | 24.2 | 205.8 | 21.99 |
| PI + AW, DT ($N$=20) | 0.52 | 0.181 | 26.1 | 171.3 | 10.42 |
| 6-St. LQR, DT ($N$=20) | 0.16 | 0.001 | 21.2 | 128.5 | 14.44 |
| PPO ref. (50 Hz) ($N$=20) | 3.20 | 4.82 | 3.15 | 54.2 | 18.40 |
| **ACC** ($N$=10) | **20.00** | **0.060** | **0.010** | **0.18** | **3.28** |

LQR / LQR+AW / 6-St. use the PID servo; the DT rows command torque directly. The three servo rows are identical because that servo's response dominates before the gain differential between them can act. Survival time is deterministic on all six classical rows (±0.00 s at $N$=20, seeds $\{0, 42, 123\}$).

reference stands longer at a single height but falls in 85 % of multi-height command transitions.

*What a* 100 % *fall rate does and does not claim.* The axis that ends the episodes fixes the scope. None of these controllers fails at the task LQR is credited with on Ascento (Klemm et al., 2019) or DIABLO (D. Liu et al., 2024): the servo variants hold pitch to 0.006° RMS and the full-state design of §5.4.1 holds 0.12° for the full 20 s. What terminates every episode is *roll*, on a morphology carrying two lateral degrees of freedom per leg that those platforms lack—and the published designs do not ask a single LQR to close that axis either, DIABLO running separate yaw, split-angle, height and roll loops around its balance controller (D. Liu et al., 2024). These rows therefore support a narrow morphological claim—reduced-order LQR of the form the wheeled-biped literature uses does not transfer to a 10-DOF articulated-leg plant *as a single controller for all axes*—not a claim that LQR cannot balance a wheeled biped. The same distinction explains why ACC's proportional-only rung L0 stands 10/10 (Table 1) where an optimal design does not: L0 retains the dedicated roll, yaw and posture channels that no row of Table 3 has. The comparison is one linearisation for all axes against per-axis structure, not optimal-versus-hand-tuned: remove the structure and add optimality and the plant falls in 0.32 s; keep the structure and strip the rest back to a proportional term and it stands—at 17.461 mm repeatability and a 69.7 mm limit cycle, the gap the remaining terms buy.

### *5.4.1. A full-state LQR designed on the plant itself*

All six baselines are designed on a reduced model, confounding method with model. The control experiment that separates them derives a regulator from the plant directly: a damped Gauss–Newton solve for a true static equilibrium at the commanded height, central differences on the whole held-input control step of the complete 16-DOF, 10-actuator MuJoCo model for the transition Jacobians, and a discrete-time Riccati solve on the 30 states left after removing the two cyclic wheel angles, with Bryson's rule fixing $\mathbf{Q}$, $\mathbf{R}$ and a uniform control weight $r$ swept over six decades. It is sound on every diagnostic that bears on it—residual unbalanced force 0.428 N against 79.46 N of body weight, $\mathbf{A}$ of full rank 30 and PBH-stabilisable, Riccati relative residual $\sim 10^{-14}$, closed-loop max $|\lambda|$=0.9911, and $\kappa(\mathbf{A})\approx 10^{10}$ a 50 Hz artifact ($2.6\times 10^{7}$ at 100 Hz) from contracting contact modes that obstructs neither solve nor feedback. What the sweep shows is a slow divergence a short horizon hides: over 20 s the design looks successful ($r$=1000 at 0.65 m: 0 % falls, 11.0 mm height RMSE), while at 60 s every configuration falls in 100 % of episodes at 22.2 – 57.0 s, ended by planar drift growing monotonically to 0.43 – 0.63 m rather than by attitude. Nor does one weight cover the band: $r$=100 stands 20 s at 0.60 m and 0.69 m but falls at 0.65 m, $r$=1000 does the reverse, under random height commands the fall rate runs 45 – 80 %, and the maximum recoverable push is 10.0 – 20.2 N against ACC's $F_{min}$=82 N. Full-state optimal linear feedback about a single equilibrium therefore buys two orders of magnitude of survival over the reduced-order designs and still does not close the problem. ACC, held to the same lengthened horizon, returns the opposite verdict—10/10 at 300 s, 4/4 at 1800 s, residual drift 0.0004 mm/min against this design's 0.43 – 0.63 m before falling (Section 5.1.4).

### *5.5. Whole-Body Control Baselines*

Whether a tuned whole-body controller could match ACC was tested over 225 scenarios spanning fixed-height balance, height transitions and push recovery at 20 – 120 N under cloned initial conditions (481 000 control steps, no gain retuning), with the task-priority quadratic-program WBC verified beforehand against MuJoCo's `mj_jac`, four independently selectable task weightings and the CoM-$z$ convention of §3.1. As the *primary* balance controller it falls in **225 of 225** scenarios, where ACC falls in none: it holds the robot upright for 0.570 s and fails *laterally* (roll RMS 93.45° against ACC's 3.71°). Neither compute nor task weighting is the constraint—the QP solve costs 0.169 ms mean (OSQP (Stellato et al., 2020))—the mechanism being that the posture task is a velocity damper rather than a position tracker, its reference rebuilt from the measured joint vector every step, so it collapses to $-k_{d,\text{posture}}\dot{\mathbf{q}}$ and can slow a postural drift but not reverse one. Blended instead as an assist on ACC, an ungated $\alpha$=0.25 never falls but degrades pitch RMS 0.52°→1.295° and planar drift 2.0×; gated, the admitted authority is $\bar{\alpha}$=0.0061 on the one variant where it opens at all. Roll is the exception, improving to 3.20° because the WBC closes it with a centralised orientation task where ACC uses decentralised hip-roll PD—a credible direction for ACC's weakest axis, the 98.2 N mean lateral bearing of Sec. 5.2, but not through torque-level blending with a posture stack that cannot hold a posture. The complete 225-scenario matrix is in the extended version.

### 5.6. Architectural Analysis

*Robustness to sensor noise and actuator delay.* A factorial sweep over four noise levels and three delay levels ($N$=20 per cell, 240 trials) bounds ACC on both axes at once (Table 4). One control period of delay is benign for stability and expensive for performance: at 10 ms there are zero falls on clean and low noise, but $F_{\max}$ nearly halves and idle RMS rises 5.6×. Three control periods is a different regime, the platform falling 20 of 20 times on *clean* sensing at 30 ms. Noise alone is survivable to the medium level; the *conjunction* is not—medium noise with 10 ms of delay produces the only sub-catastrophic falls anywhere in the grid. High noise is unrecoverable at every delay including zero, because the asymmetric envelope cannot distinguish noise from motion—the same mechanism behind the design's sharpest single sensitivity, idle repeatability degrading 71-fold under medium noise at zero delay as the envelope gate releases and re-engages the integral. ACC is therefore specified against a filtered velocity estimate, a requirement on the state estimator rather than a limit on the controller.

*Resolving the delay axis.* The grid quantises delay to the control period, so the knee was located separately with the transport delay applied per *physics* substep (2 ms at 500 Hz), the clean-sensing harness being deterministic. It lies between 6 ms and 8 ms: $F_{\max}$ is flat to within 2.4 % out to 6 ms (93.2 – 95.5 N), then collapses 45 % to 51.0 N and declines monotonically to the 10 N search floor by 20 ms—a genuine loss of margin rather than a narrow phase-alignment notch, and the 6.6 – 8 ms end-to-end budget estimated in §6 lands inside that bracket rather than clear of it. Losing the push margin and losing the platform are different thresholds: the standing threshold is 14 – 16 ms, station held through 14 ms at 10.84 mm idle lateral RMS against 0.43 mm undelayed. The knee is not an exhausted margin: holding the delay at 8 ms and sweeping one profile constant at a time gives a sharp *local dip*, the shipped value of each of four constants tested scoring 51.0 N while nearly every neighbouring value recovers 78 – 93 N. Carrying the largest single-knob change forward—sagittal velocity damping doubled to 45 Nm/(m/s), denoted ACC-DH—leaves the controller identical at 0 – 2 ms and moves the knee outward to 8 – 10 ms (51.0 → 93.2 N at 8 ms) for +1.4 % idle lateral RMS. ACC-DH is reported as an evaluation variant and not adopted, since the constant it changes also governs driving acceleration and settling.

*Gate sensitivity and rate-limit invariance.* Finite-difference gradients give a stark hierarchy: the release coefficient $\alpha_r$ dominates (sensitivity 220.4, 20×

**Table 4.** Robustness to sensor noise and actuator delay, $N$=20 per cell. Falls: trials terminated by a fall, out of $N$. Idle RMS: lateral (world $x$) CoM std. over 20 s after a 3 s settle, *surviving trials only*. $F_{\max}$: binary-search lateral (world $+x$) push, ±5 N tolerance, search floor 10 N. Delay is quantised to the control period here; it is resolved five times more finely in the text.

| Condition | Falls | Idle RMS (mm) | | $F_{\max}$ (N) | |
|---|---|---|---|---|---|
| | | Mean | ±Std | Mean | ±Std |
| Clean, 0 ms delay | 0/20 | 0.43 | 0.06 | 95.7 | 0.8 |
| Clean, 10 ms delay | 0/20 | 2.43 | 0.21 | 50.5 | 7.9 |
| Clean, 30 ms delay | **20/20** | — (all fell) | | — | |
| Low noise, 0 ms delay | 0/20 | 5.86 | 2.53 | 96.0 | 1.0 |
| Low noise, 10 ms delay | 0/20 | 10.66 | 3.70 | 51.3 | 8.9 |
| Low noise, 30 ms delay | **20/20** | — (all fell) | | — | |
| Med noise, 0 ms delay | 0/20 | 30.82 | 12.13 | 93.5 | 5.2 |
| Med noise, 10 ms delay | **2/20** | 35.38 | 21.63 | 51.3 | 13.4 |
| Med noise, 30 ms delay | **20/20** | — (all fell) | | — | |
| High noise, all three delays | 20/20 | — (all fell) | | — | |

Noise (1-$\sigma$): low = 0.01 rad/s angular velocity, 0.01 m/s$^2$ gravity, 0.001 rad joint position; med = 5× low; high = 10× low. Delay is a transport lag on the actuator command. High noise fails identically at all three delays, so those rows are merged. Because the idle column averages survivors only, the medium-noise 10 ms row is mildly optimistic and its falls column is the primary metric. Produced by `scripts/collect_robustness_sweep.py`.

the next and 130× the pitch gates), while the pitch thresholds are insensitive safety interlocks. The three gate failure modes map onto measured rows of Table 2: stuck-at-1 causes windup and a fall (S4), stuck-at-0 the P-only limit cycle (L0), stuck-half-open the parametric resonance of S5. The 400 Nm/s slew limit does not set the push ceiling: over an 8× span lateral $F_{max}$ is 92.5±1.1 N at 100 Nm/s and 94.4±0.0 N at 200, 400 and 800 Nm/s, while during a 90 N push the commanded slew saturates exactly at each bound—so the ceiling is architectural, not actuator-limited.

*Compound disturbance.* A commanded height transition and a push compete for the same leg-torque budget, so the standard lateral impulse is applied at the midpoint of a 1 s height ramp of size $\Delta h$ about the 0.404 m nominal, with $\Delta h$=0 as the matched static control ($N$=10 per row). Inside the calibrated band a simultaneous height command is nearly free (+5 cm costs 5.1 % of push margin; −5 cm is *better* than static, +6.5 %), and the degradation outside it is one-sided: −10 cm costs nothing measurable ($p$=0.99) while +10 cm costs 68.7 %, since rising drives the knee toward extension—where the leg Jacobian is worst conditioned—while raising the CoM; re-parameterising the same posture family so achieved height matches the command closes 35 % of that deficit ($p=1.5 \times 10^{-4}$). Sequential disturbance is absorbed cleanly: a 90 N lateral impulse followed 2 s later, before the first ringdown has decayed, by a 60 N counter-directed impulse is survived **10/10** at episode peak pitch 11.89±0.51°—what $\tau_{release}\approx 1.5$ s is designed to produce, the damping boost still partly engaged when the second impulse arrives.

### 5.7. Local Stability Certificate at the Standing Fixed Point

ACC's gates make it a switched system, of the kind partitioned by discrete switching surfaces in the hybrid-systems literature (Liberzon, 2003) and softened by hysteresis in supervisory switching logic (Hespanha et al., 2003). Everything reported so far is behavioral evidence rather than a stability statement; this section supplies one.

*At quiet stance the switching is inactive, by a measured margin.* Over the last 20 s of a 60 s settle, 19 of the 21 gates the controller exports sit at exactly 0 or exactly 1 at every control step: bit-exact saturation, not approximate. The proximity gate's argument $|e_{sag}|$ peaks at 24.8 mm against its 50 mm knee (16.1 – 33.2 mm across the five height variants), and the envelope gate's argument peaks at $1.5\times10^{-7}$ m/s against its 0.25 m/s knee, a margin of $1.7\times10^{6}$. The two gates that do sit inside their bands—anti-twist at ≈0.53, centering at ≈0.986—are ordinary smooth feedback, a smoothstep being $C^{\infty}$ in the interior of its band. In a neighbourhood of the fixed point every branch selection is therefore frozen, the closed loop is a single smooth map, and Lyapunov's indirect method applies: chattering is avoided not because the gates are $C^1$ but because the operating point stands the above margins away from every switching surface.

*The one switch that does fire is a contraction on both branches.* The asymmetric envelope of Eq. (14) selects $\alpha \in \{0.35, 0.0067\}$ on the sign of $|v_{sag}| - \mathrm{EMA}_{t-1}$, which in quiet stance flips at essentially every step, so the composition is only $C^0$ *in time*. For a fixed input level the tracking error $e_t = \mathrm{EMA}_t - |v_{sag}|$ obeys $e_{t+1} = (1-\alpha_t)e_t$, so $|e_{t+1}| \leq 0.9933\,|e_t|$ under *either* branch: $V(e) = e^2$ is a common Lyapunov function for the switched pair, and the envelope is globally exponentially stable under arbitrary switching, worst-case rate set by the slow branch ($\tau$=1.49 s). Switching among individually stable subsystems therefore cannot destabilize the whole (Liberzon, 2003) and no dwell-time or hysteresis condition is required (Hespanha et al., 2003)—and in any case the discontinuity reaches the loop only through a gate that is flat (≡ 1) across the entire quiet band, so its switching cannot modulate loop gain in the certified region.

*The linearized closed loop.* We central-difference the whole 100 Hz control step—control law plus five physics substeps—about the settled state, over an augmented vector of 57 coordinates: 30 plant (16 velocities and 14 configuration tangents, the two cyclic wheel angles removed as in Section 5.4.1), 3 of estimator memory and 24 of controller memory, with the adaptive-bias trim held at its converged value so that the one-step map is time-invariant rather than 300-periodic. At all five heights the spectrum has exactly two eigenvalues on the unit circle, none outside it, and a reduced spectral radius of 0.997 87 – 0.998 18, a slowest stable mode of 4.7 – 5.5 s (Table 5). The classification is not a threshold judgement—the gap from the unit circle down to the next mode is $1.8\times10^{-3}$, four orders of magnitude above the $10^{-7}$ sensitivity of the spectrum to the differencing step.

*Both centre directions are identified and bounded by measurement.* The first is lateral translation: a differential-drive base produces no lateral force to first order, a non-holonomic centre direction at $|\lambda|$=1 to $10^{-12}$. The second is deliberate—the heading channel carries an explicit deadband, ss($|e_{\text{yaw}}|$; 0.02 rad, 0.08 rad), measured at exactly 0 here, so absolute yaw enters no control law while the error stays inside ±1.15°. Neither is left to the linearization: over 300 s of the full nonlinear simulator a 1.000 mm lateral offset returns 0.997 mm and a 0.500° yaw offset creeps to 0.565° at $+4.8\times10^{-5}$ °/s, needing 3.3 h to reach the deadband edge at which heading control re-engages.

*Cross-validation, and scope.* Random plant perturbations at $10^{-4}$, $10^{-3}$ and $10^{-2}$ decay in the full nonlinear simulator—adaptive bias live, nothing frozen—at $\rho$=0.99521±0.00237, 0.99513±0.00117 and 0.99547±0.00355, against the 0.995 30 predicted for the anchor-integral mode ($\tau$=2.12 s, 0.033 Hz) that a generic perturbation excites: agreement to $2\times10^{-4}$ over two decades of amplitude. The certificate is local and on the nominal model, at zero delay and clean sensing; it estimates no region of attraction, and the push results of Section 5.2 lie outside it by construction, a push being what drives the gates off their flat regions. It establishes that the point ACC parks at is exponentially attracting in 55 of 57 directions and neutral in the other two, both named, mechanistically explained and separately bounded.

**Table 5.** Local stability certificate at the standing fixed point, across the five calibrated height variants. Central-difference linearization of the 100 Hz closed-loop step after a 60 s settle; "Flat" counts exported gates bit-exactly saturated over the last 20 s. $\rho_{\text{red}}$ excludes the two centre directions, bounded separately by 300 s nonlinear probes.

| Variant | CoM $z$ (m) | Flat | Ctr. | Unst. | $\rho_{\text{red}}$ | $\tau_{\text{slow}}$ (s) |
|---|---|---|---|---|---|---|
| Nominal | 0.4069 | 19/21 | 2 | 0 | 0.99818 | 5.49 |
| High small | 0.4164 | 18/21 | 2 | 0 | 0.99787 | 4.70 |
| High tiny | 0.4128 | 19/21 | 2 | 0 | 0.99817 | 5.46 |
| Low small | 0.3973 | 19/21 | 2 | 0 | 0.99813 | 5.34 |
| Low tiny | 0.4009 | 19/21 | 2 | 0 | 0.99815 | 5.41 |

Produced by `scripts/acc_stability_certificate.py`; full spectra, gate traces and probe trajectories in the released output. Spectra are insensitive to the differencing step at the $10^{-7}$ level.

## 6. Discussion and Conclusion

Every gate in ACC was added after tracing a measured failure mode to its root cause: a bistable symmetric envelope, global integral windup, parametric pumping of a scheduled stiffness, hard-leash oscillation and self-referential stiffness. All thresholds were frozen before the ablation sweep, so Table 2 scores a fixed specification rather than a tuned one, and because the gating variables are physical quantities the controller runs without training and is hypothesised, pending hardware validation, to carry a reduced sim-to-real gap (Grimminger et al., 2020).

**Statistical power.** Welch's $t$-test at Bonferroni-adjusted $\alpha$=0.007 (7 comparisons) makes L0→L1 and L2.5→L3 significant ($p$<0.001; Cohen's $d$=5.4 and 4.1); $N$=10 success rates carry a Clopper–Pearson 95% CI of [0.69, 1.00], push ablations resolve ~2 N, and no claim rests on less.

**Limitations.** (1) *Simulation only.* Hardware validation remains future work. (2) *Deployment timing is the binding constraint.* ACC as tuned requires a 100 Hz loop and a latency clearing the 6 – 8 ms push-margin cliff of §5.6. Rate independence is not claimed: at 50 Hz with recomputed coefficients the platform does not stand, and the cause—plant-mode excitation or phase lag at $\Delta t$=20 ms—is unresolved. (3) *The calibrated height envelope is non-uniform.* A +10 cm height command during a push costs 69 % of $F_{\max}$, descent being free at every amplitude. (4) *The learning comparison is preliminary, and one classical family remains untested.* The model-free PPO run (5.4M steps, one seed, no domain randomisation (Tobin et al., 2017) or hyperparameter search) is a difficulty probe bounding nothing about RL here; the comparison this controller is built for—residual learning *over* the prior—is not run. A tuned convex MPC over the linearisation of §5.4.1 is the natural next comparator. (5) *Generality is undemonstrated.* Every result is for one 10-DOF platform with one set of gains.

**Deployment risks.** ACC needs only raw torque, a 400 Nm/s rate limit, an IMU and encoders. The binding figure is the 6 – 8 ms latency cliff in push margin (§5.6); the stability threshold is twice as far out (station held through 14 ms), the ordering the architecture predicts. An estimated non-RTOS budget totals ≈8 ms: sensor readout ~1 ms, state estimation ~2 ms, actuation ~3.5 ms—all assumed—plus a *measured* ≤1.5 ms for the control step (1.27 – 1.52 ms p99 across five repeats of 5000 steps at the quiet-stance operating point).

Host wall-clock is nevertheless the wrong quantity to extrapolate to an embedded target; the reproducible one is the operation count: 27 664 floating-point operations and 159 transcendentals per step over 1001 scalars, identical on every repeat, with no iterative solve, no matrix factorization and no data-dependent control flow—every gate of §4.3 is a branchless select—so worst-case execution time equals average-case, which a QP-based controller cannot offer. A 480 MHz Cortex-M7 retires that in ~0.07 ms and a 100 MHz Cortex-M4 in 0.34 ms, so compute is the one budget line that shrinks on the target: ~95 % of the measured 1.2 ms sits outside the compiled kernel, in CoM and support-polygon work plus Python dispatch that a C port deletes. Charged at 0.1 ms the estimate is 6.6 ms: against the 6 – 8 ms cliff the instrumented figure sits at the top of the bracket and the projected one at its bottom, so readout and actuation decide the matter, not compute. Two mitigations compose—a real-time OS recovers 2 – 4 ms (loop at 2.6 – 4.6 ms projected), and doubling the sagittal velocity damping moves the knee to 8 – 10 ms without touching the schedule. Hardware repeatability is projected at 1 – 3 mm rather than 0.294 mm.[2]

**Conclusion.** We presented ACC, a two-channel torque assembly in which a PD balance core is augmented by a proximity-gated anchor integral, achieving 0.294±0.015 mm idle repeatability on the sagittal axis—a 59× improvement over the proportional-only baseline ($N$=10, Table 1)—with omnidirectional push survival from $F_{\min}$=82 N to a 107 N median. Absolute accuracy is separate: the robot parks 27.0 mm from commanded home and holds that standoff to 0.009 mm, which a closed-form model predicts as the static price of a bounded integrator memory to within 0.45 %. Six classical baselines and a full-state LQR derived from the plant itself all fall, and integral action does not prevent it. ACC tolerates moderate noise, is insensitive to the slew bound over an 8× span, and extends to contact loss and per-leg terrain without training; its sharpest deployment constraint is actuator latency. The contribution is a *structured classical prior*: an interpretable controller whose gating encodes the plant's measured failure modes. Every claim for it is a directly measured property of the controller alone—the mechanism decomposition, the push-envelope chirality, the closed-form parking offset, the stability certificate of §5.7—so the results do not wait on the residual stack the prior was built to serve. That stack is future work, alongside hardware validation, a resourced learning comparison, and conditional activation for locomotion.

[2]Produced by `scripts/measure_control_step_cost.py`.

## Code and Data Availability

Controller, MuJoCo model, evaluation scripts and raw results are at `https://github.com/thuong180702/Wheeled-bipedal-robot-simulation`; each table's note names its script. The extended version there carries the full derivations, WBC details and parking-offset model.

## 7. Notes on contributors


*Van Thuong Nguyen* is an independent researcher based in Viet Nam. His research interests include classical and learning-based control for wheeled bipedal platforms, whole-body control, and structured model-based controllers used as action priors for residual reinforcement learning.


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